

Eight-Point Exact-Pressure Optimization for Simple Zeros of the Riemann Zeta Function

AMTOPA

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Abstract

We record a computer-assisted research-draft refinement of the positioned-pressure method for simple zeros of the Riemann zeta function on the critical line. The refinement retains exact shifted-block pressure accounting and a rigorous 15-term analytic window, but enlarges the local certificate from seven points to eight points and optimizes exact pair/position-pressure weights under the span-capacity constraints used by the predecessor deduction. A seven-dimensional outward-rounded interval computation proves the resulting local inequality. Subject to the same imported analytic interface and finite-dimensional Gram profile as the predecessor lineage, the certified numerical inputs yield

$$\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} \geq 0.67337819547776013737 \dots,$$

i.e. 67.3378195478% before conservative decimal truncation. This manuscript is a research draft, not a peer-reviewed theorem and not a proof of the Riemann hypothesis.

1 Imported analytic interface

Let $S = N_0^s(T, 2T)$ and $N = N(T, 2T)$. We import

$$S \geq HN + \Delta(M) - o(N)$$

and the finite- m Gram profile

$$h_m(E) = \begin{cases} E, & 0 \leq E \leq m/(m-1), \\ E/m + 2\sqrt{(m-1)E/m} - 1, & E \geq m/(m-1). \end{cases}$$

These ingredients belong to the Anthropic / `trmdy` / `sxuff` lineage and are not reproved here.

2 Analytic window

We use the rational 15-term window

$$v(s) = \sum_{j=0}^{14} c_j \cos(\omega_j s), \quad \omega_0 = \sqrt{2}, \quad \omega_j = 2j\pi \quad (1 \leq j \leq 14),$$

with common denominator 10^9 and numerator vector

$$(1000000000, 8629738, -10085378, 1746803, 1125700, -2203905, 1904615, \\ -4559603, 7930665, -3022627, 2165339, 398121, -255934, 188899, -148305).$$

Direct interval arithmetic verifies

$$H_{\text{cert}} = \frac{3361669433}{5000000000} = 0.6723338866 < H(v).$$

3 Eight-point local functional

Let $g_1, \dots, g_7 \geq 0$ and $y_0 = 0$, $y_j = g_1 + \dots + g_j$. We use

$$F_8(g) = \sum_{r=1}^7 b_r g_r + \sum_{0 \leq i < j \leq 7} a_{ij} W(y_j - y_i).$$

The exact pair weights are recorded in `candidate.json`. They are nonnegative and satisfy the span identities

$$\sum_{i=0}^{7-s} a_{i,i+s} = 2 \quad (1 \leq s \leq 7).$$

The position-pressure coefficients are

$$\frac{1}{2300000000}(474488, 829921, 1102531, 1186120, 1102531, 829921, 474488),$$

so

$$B = \sum_{r=1}^7 b_r = \frac{3}{1150}.$$

These are exactly the pair-capacity and total-pressure facts used in the predecessor local-to-global deduction; the counting is dimension-generic and replacing six gaps by seven gives $m - 7$ translated local windows in an m -point block.

4 Seven-dimensional interval certificate

A reflection-symmetric trust-region optimization gave the exact rational local weights. Unscreened floating-point stress over the relevant resonance families found a lowest basin near

$$0.005482799831021096 \dots$$

We certify the smaller rational target

$$\varepsilon_8 = \frac{2741}{500000} = 0.005482.$$

The rigorous verifier uses 4000 interval cells per normalized unit, 50 decimal digits in the outward-rounded table builder, and C++ compiled with `-ffp-contract=off`. It reports

$$\begin{aligned} \text{nodes} &= 56348888, & \text{pruned} &= 28174468, \\ \text{splits} &= 28174420, & \text{convex} &= 21755661, \\ \text{tangent} &= 9522833, & \text{max depth} &= 83. \end{aligned}$$

The output is `VERIFIED=true`. Exact table hashes, workflow run 31594502822, and artifact 9141284166 are stored in the repository metadata.

5 Exact shifted-pressure averaging

For an m -point block there are $m - 7$ translated eight-point windows. Summing their local inequalities yields the threshold

$$A_m = \varepsilon_8(m - 7).$$

The span-capacity identities ensure that the summed pair contribution is dominated by the predecessor block pair energy, while exact shifted averaging gives pressure charge

$$(m - 7)B.$$

Set

$$R_m = h_m(A_m), \quad \eta_m = \frac{R_m}{A_m}.$$

The block-defect estimate becomes

$$\Delta(M) \geq \frac{R_m}{m}S - \frac{\eta_m B(m-7)}{m}N - o(N),$$

and therefore

$$\boxed{\frac{S}{N} \geq \frac{mH_{\text{cert}} - \eta_m B(m-7)}{m - R_m} - o(1).}$$

6 Final arithmetic

With

$$H_{\text{cert}} = 0.6723338866, \quad \varepsilon_8 = 0.005482, \quad B = 3/1150,$$

the integer scan selects $m = 204$. Then

$$A_{204} = 1.079954,$$

$$R_{204} = 1.0786101804671024521213591572698778 \dots,$$

and

$$\eta_{204} = 0.9987556696554690774990038068935 \dots$$

Hence

$$\frac{S}{N} \geq 0.673378195477760137371654170550 \dots - o(1),$$

and safely

$$\boxed{\liminf_{T \rightarrow \infty} \frac{N_0^s(T, 2T)}{N(T, 2T)} > 0.6733781954.}$$

7 Trust boundary and status

The interval window bound, exact pair-span identities, exact pressure total, seven-dimensional local certificate, shifted-pressure bookkeeping, and final arithmetic are directly checkable from this repository. The explicit-formula / trace interface and finite- m Gram profile remain imported from prior work. Independent reproduction of the new seven-dimensional certificate is requested; accordingly, the percentage should be read as an interval-certified research-draft candidate rather than a publication-ready theorem.

Reproducibility

The repository contains exact current parameters in `candidate.json`, verifier evidence in `certificates/`, arithmetic and interval code in `src/`, and superseded records in `archive/`. The repository follows a single-main workflow.